IBM Quantum Composer - Circuit Modifications Workshop

Student Lab Handout

Session Overview

Drive your own discovery in IBM Quantum Composer. Working in pairs, first recreate the basic circuits, then experiment with modifications to observe key effects. This hands-on approach builds circuit intuition through structured trial-and-error and documentation.

Session Objectives

- Accurately recreate all-zero, X-gate, Bell, interference, and GHZ circuits
- Experiment with modifications (gate order, added measurements) and predict outcomes
- Record histograms, QASM snippets, and observations in a structured format
- Build debugging skills for non-ideal results (queue noise)

Timed Agenda (75 minutes)

Time	Activity
0:00-0:10	Login/setup check; M1 recap; distribute handout
0:10-0:50	Paired circuit reproduction + modifications
0:50-1:08	3-4 pairs share one modification + histogram
1:08-1:15	Notebook wrap-up, submit, Lab 4 preview

Workshop Tasks - quantum.ibm.com/composer

Settings: Simulator (1024 shots). Save as "Lab3_CircuitX_vY"; download CSV/screenshots.
Convention: In IBM Quantum Composer, the first qubit (q_0) is on the right.

Task 1: Reproduce Basis Circuits (15 min)

Build the following circuits and verify histograms.

Circuit	$\|0\rangle$%	$\|1\rangle$%	$\|00\rangle$%	$\|11\rangle$%	**Job ID**	✓
1A: All-Zero						
1B: X-Flip						
1C: Bell						

Task 2: Single-Qubit Modifications (10 min)

2a. $|0\rangle \rightarrow \mathrm{H} \rightarrow \mathrm{X} \rightarrow \mathrm{M}$

2b. $|0\rangle \rightarrow \mathrm{X} \rightarrow \mathrm{H} \rightarrow \mathrm{M}$

2c. $|0\rangle \rightarrow \mathrm{X} \rightarrow \mathrm{M} \rightarrow \mathrm{H} \rightarrow \mathrm{M}$

Part	**Circuit**	**Predict**	**Actual**	**Why?**
(a)	$\|0\rangle \rightarrow \mathrm{H} \rightarrow \mathrm{X} \rightarrow \mathrm{M}$	_____	_____	_____
(b)	$\|0\rangle \rightarrow \mathrm{X} \rightarrow \mathrm{H} \rightarrow \mathrm{M}$	_____	_____	_____
(c)	$\|0\rangle \rightarrow \mathrm{X} \rightarrow \mathrm{M} \rightarrow \mathrm{H} \rightarrow \mathrm{M}$	_____	_____	_____

Table 1: Prediction, actual results, and reasoning for circuits (a), (b) and (c)

Task 3: Bell Circuit Variations (15 min)

Test three modifications to the standard Bell circuit (H on q_0, $\mathrm{CNOT}_{0\rightarrow1}$).
Which ones preserve entanglement? Check if single-qubit marginals remain uniform 50/50.

3a. Swap H and CNOT order (product state?):

$$|00\rangle \xrightarrow{\mathrm{CNOT}_{0,1}} |00\rangle \xrightarrow{\text{H on } q_0} \frac{|00\rangle + |10\rangle}{\sqrt{2}} = |+0\rangle$$

Note: $|+0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ is exactly the same as $|+\rangle \otimes |0\rangle$
Expected histogram: $|00\rangle$ and $|10\rangle$ only (50/50).

Expected Results Table:

Circuit	$\|00\rangle$%	$\|10\rangle$%	**Others?**	**Job ID**	✓
Swap H/CNOT					

3b. Measure q_0 first (classical correlation?):

Add measurement on q_0 before q_1. Does early measurement destroy entanglement?
Expected histogram: $|00\rangle$, $|11\rangle$ only? Or all combinations?

Expected Results Table:

Circuit	$\|00\rangle$%	$\|11\rangle$%	**Others?**	**Job ID**	✓
Measure q_0 first					

3c. Z on q_1 before CNOT (phase flip?):

$$|00\rangle \xrightarrow{\text{H on } q_0} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\text{Z on } q_1} \frac{|00\rangle - |10\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}_{0,1}} \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

Expected histogram: $|00\rangle$ and $|11\rangle$ only (50/50) - still entangled!

Expected Results Table:

Circuit	$\|00\rangle$%	$\|11\rangle$%	**Others?**	**Job ID**	✓
Z on q_1 pre-CNOT					

How to Check Single-Qubit Marginals (50/50 Uniformity)

The **marginal** of a qubit is its individual probability distribution, *ignoring* the other qubit(s). Sum over all outcomes of other qubits.

Step 1: Record histogram from IBM Quantum Composer

Example: $|00\rangle$: 512, $|11\rangle$: 512 (1024 total shots)

Step 2: Calculate marginal probabilities

For q_0 (first qubit):

$$P(q_0 = 0) = \frac{|00\rangle + |01\rangle}{\text{total}} = \frac{512 + 0}{1024} = 50\%$$

$$P(q_0 = 1) = \frac{|10\rangle + |11\rangle}{\text{total}} = \frac{0 + 512}{1024} = 50\%$$

For q_1 (second qubit):

$$P(q_1 = 0) = \frac{|00\rangle + |10\rangle}{\text{total}} = \frac{512 + 0}{1024} = 50\%$$

$$P(q_1 = 1) = \frac{|01\rangle + |11\rangle}{\text{total}} = \frac{0 + 512}{1024} = 50\%$$

Step 3: Entanglement Signatures

- **ENTANGLED:** Both marginals 50/50 **AND** only $|00\rangle$, $|11\rangle$
- **PRODUCT STATE:** At least one marginal $\neq$ 50/50

Task 4: 3-qubit GHZ State (10 min)

Extend Bell state to 3-qubit GHZ: all entangled together.

Build on 3 qubits (q_0,q_1,q_2):

$$|000\rangle \xrightarrow{H \text{ on } q_0} \frac{|000\rangle + |100\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}_{0,1}} \frac{|000\rangle + |110\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}_{1,2}} \frac{|000\rangle + |111\rangle}{\sqrt{2}}$$

Expected histogram: $|000\rangle$ and $|111\rangle$ only (50/50).

Expected Results Table:

Circuit	$\|000\rangle$%	$\|111\rangle$%	**Others?**	**Job ID**	✓
GHZ State					

Completion Checklist

- ☐ All 3 basis circuits reproduced
- ☐ 3 single-qubit modifications completed
- ☐ 3 Bell variations tested
- ☐ 3-qubit GHZ state ($|000\rangle$+$|111\rangle$)
- ☐ Key histograms screenshot
- ☐ Predictions vs. actual documented
- ☐ Submitted by **February 23**

Additional Questions

- Which modifications showed **entanglement**? ($|00\rangle + |11\rangle$ *and* $|000\rangle + |111\rangle$)
- Why does **gate order** matter? (Interference vs. classical probability)
- What breaks **correlations**? (Early measurement, wrong CNOT order)

Lab 4: Formal lab reports + Qiskit coding!